

Blended learning and augmented employability: a multi-stakeholder perspective of the micro-credentialing ecosystem in higher education

Blended
learning in
higher
education

1021

Received 5 December 2022

Revised 10 May 2023

15 December 2023

Accepted 6 March 2024

Himani Sharma and Varsha Jain

Mudra Institute of Communications, Ahmedabad, India

Emmanuel Mogaji

*School of Management and Marketing, University of Greenwich,
London, UK and*

Keele Business School, Keele University, Keele, UK, and

Anantha S. Babbilid

Texas A&M University – Corpus Christi, Corpus Christi, Texas, USA

Abstract

Purpose – Proponents of micro-credentials envision them as vehicles for upskilling or re-skilling individuals. The study examines how integrating micro-credentials in the higher education ecosystem enhances employability. It aims to offer insights from the perspective of stakeholders who may benefit from these credentials at an institutional or individual level.

Design/methodology/approach – Online in-depth interviews are conducted with 65 participants from India, Nigeria, the United Arab Emirates and the United Kingdom to explore how micro-credentials can be a valuable addition to the higher education ecosystem. A multi-stakeholder approach is adopted to collect data.

Findings – The analysis highlights two possible methods of integrating micro-credentials into the higher education ecosystem. First, micro-credentials-driven courses can be offered using a blended approach that provides a flexible learning path. Second, there is also the possibility of wide-scale integration of micro-credentials as an outcome of standalone online programs. However, the effectiveness of such programs is driven by enablers like student profiles, standardization and the dynamics of the labor market. Finally, the study stipulates that micro-credentials can enhance employability.

Originality/value – The study's findings suggest that, for successful integration of micro-credentials, an operational understanding of micro-credentials, their enablers and strategic deliberation are critical in higher education. Institutions must identify the determinants, address technological limitations and select a suitable delivery mode to accelerate integration. However, micro-credentials can augment employability, considering the increasing emphasis on lifelong learning. An overview of the findings is presented through a comprehensive framework.

Keywords Micro-credentials, Higher education, Blended approach (online and offline), Employability

Paper type Research paper

1. Introduction

In recent years, the landscape of higher education has witnessed a transformative shift, with the emergence of micro-credentials as a pivotal component in institutions pushing beyond the conventional boundaries of learning (Pirkkalainen *et al.*, 2023). This phenomenon, although pre-existing, has garnered increased attention due to the growing demand for competency-based skills in today's job market (McGreal *et al.*, 2022). The heightened interest in



International Journal of
Educational Management
Vol. 38 No. 4, 2024
pp. 1021-1044

© Emerald Publishing Limited
0951-354X
DOI 10.1108/IJEM-12-2022-0497

This paper forms part of a special section “Micro-credential as a Digital Enabler for Higher Education Ecosystems”, guest edited by Muhammad Ali, Syed Ali Raza, Chin Hong Puah and Wasim Qazi.

micro-credentials aligns with the strategic focus of higher education institutions on personalized student learning, manifested through partnerships with diverse online learning platforms to deliver skill-driven courses (Brown *et al.*, 2021). The Organization for Economic Cooperation and Development (OECD) recognizes the expanding significance of “micro-credentials” in education and training, yet their seamless integration into the higher education ecosystem remains nuanced and met with skepticism, necessitating further exploration.

A notable impediment to the widespread acceptance of micro-credentials, particularly in developing countries, is the reluctance within traditional university systems to appreciate these alternative credentials (Varadarajan *et al.*, 2023). Fear looms that adopting micro-credentials might cannibalize the existing market for traditional degrees, potentially impacting revenues. Kato *et al.* (2020) suggest that a lack of comprehensive understanding of micro-credentials across academia exacerbates such reservations. Furthermore, the absence of a unified definition for micro-credentials (Oliver, 2019) and the fragmentation within existing literature (Sankey and Selvaratnam, 2021) contribute to the prevailing skepticism.

Despite being recognized as alternative degree credentials in the form of certificates, digital badges, or nano-degrees (Ericksen, 2019), micro-credentials are not fully integrated into the standard credentialing system of higher education. Ongoing discussions surround their widespread incorporation, with European universities operationalizing micro-credentials as a sub-unit of the primary credit system (Uggeri and Barlassina, 2019). Similarly, institutions in Canada and the United States acknowledge micro-credentials’ value in addressing the shortage of highly skilled individuals in the job market (McGreal and Olcott, 2022; Young *et al.*, 2021). Nevertheless, despite the escalating importance of micro-credentials, empirical research on their integration into the higher education ecosystem is limited (Pirkkalainen *et al.*, 2023) and exploration of their acceptance in developing nations remains a largely uncharted territory (Schick, 2021).

Moreover, existing empirical research predominantly approaches micro-credentials from either the perspective of digital platforms (Kiiskilä *et al.*, 2023) or employers in organizations (Gauthier, 2020). Recognizing these gaps in the literature and considering the relative novelty of micro-credentials, this study endeavors to address the following research questions:

- RQ1. What are the enabling factors for integrating micro-credentials into the higher education ecosystem?
- RQ2. How can micro-credentials enhance employability?
- RQ3. To what extent is the blended mode of delivery suitable for integrating micro-credentials into the higher education ecosystem?

In addressing these research questions, we have employed the Learning Environment, Learning Processes and Learning Outcomes (LEPO) Framework proposed by Phillips *et al.* (2010) as our theoretical foundation. This framework provides a structured lens through which we can examine micro-credential integration into the higher education ecosystem. Our chosen methodology for this exploration is qualitative research involving interviews with various stakeholders, including students, faculty, administrators and industry professionals. This approach allows us to capture diverse perspectives and gain a comprehensive understanding of how these key actors perceive micro-credentials and their role within higher education. The study endeavors to contribute significant insights to the ongoing discourse surrounding micro-credentials. Through this qualitative exploration, we seek to shed light on the potential benefits, challenges and implications that micro-credentials hold for the continually evolving landscape of higher education. This comprehensive approach allows us to offer a nuanced understanding of the multifaceted dimensions surrounding the integration of micro-credentials, ultimately contributing to the broader conversation on the future of learning and credentialing in higher education.

2. Literature review

2.1 Overview of micro-credentials

In the context of this study, we adopt the following definition of micro-credentials: “A micro-credential is a tangible representation of the learning outcomes achieved by a learner through a brief educational experience. These outcomes are assessed against transparent standards” (Gabriel, European Commission, 2020, p. 10). Essentially, micro-credentials are certifications, in document form, that signify the successful completion of a specific course. These documents are recognized proof of a learner’s accomplishments and performance (McGreal and Olcott, 2022), validating their skills based on predefined assessment parameters.

Unlike traditional degrees, micro-credentials offer a unique flexibility; they can be stacked, paused and resumed at the learner’s convenience. Institutions providing learning programs can incorporate micro-credentials in two ways: by integrating them within the mainstream courses offered or as additional components that may or may not be stacked as accreditation units (Cirlan and Loukkola, 2020). Referred to as alternative digital credentials (A.D.C.), micro-credentials may be offered in digital format or traditional document form (Matkin, 2020). The current landscape of higher education allows institutions the flexibility to choose a suitable mode of delivery for micro-credentials, providing adaptability to meet the diverse needs of learners.

2.2 Current landscape of micro-credentials in higher education

Many online learning platforms are engaged with micro-credentials and offer some of the best possible skill-driven courses (Healy, 2021). According to a report by OECD in 2023, Coursera, a recognized online learning platform, currently offers 818 specializations, 29 master tracks and 92 professional certifications micro-credentials in collaboration with universities and industries worldwide. edX, a well-known management learning outlet, offers 62 Xseries, 18 micro-bachelors, 80 micro-masters, 344 professional certifications and 97 professional education micro-credentials. Platforms like Udacity offer 90 Nano-degrees and LinkedIn Learning offers 548 learning path micro-credentials (Kato and Gyorfi, 2023). All these platforms either independently offer these competency-driven courses or collaborate with leading universities or industry experts to facilitate the learners.

However, it is crucial to note that micro-credentials can be credit or non-credit-bearing when offered within a higher education ecosystem. For most universities and higher education institutions, micro-credentials exist outside the scope of traditional degrees despite having the potential to increase employability (Griffith, 2021). It is the discretion of the university or the institution to offer such courses to the learners. Through ongoing discussions around adopting and implementing credit bank systems in higher education, this is still a gray area and needs further attention (Chattopadhyay *et al.*, 2022). To study this convoluted area, scholars have explored the relationship between higher education and micro-credentials as going beyond transcripts and as an alternative strategic system of offering credentials in specific universities (Ghasia *et al.*, 2019). At the same time, Mischewski (2017) has discussed the role of micro-credentials in fulfilling learners’ requirements and found that personalization and flexibility were the essential requirements for learners that micro-credentials could fulfill. Pickard *et al.* (2018) later critiqued the results and discussed the inconsistent value of micro-credentials in higher education qualifications.

Along with these contrasting explanations, we found that there is also a limited focus on the enablers and mode of delivery of micro-credentials in higher education (Attri, 2019). In hindsight, it must be acknowledged that the blended approach is considered an enabler for integrating micro-credentials, especially after the COVID-19 pandemic (Kumar *et al.*, 2022). Also known as a hybrid approach, a blended approach is defined as a mode of delivery that integrates offline and online mediums (Ibrahim and Nat, 2019). It involves creating a rich

experience for learners in a conducive learning environment (Rossiter and Tynan, 2019) and has been increasingly seen as a solution (Hunt and Neves, 2020). The basic premise of the blended approach is that it can either be used to enhance flexibility in offering micro-credentials or can be used to encourage self-learning (Bordoloi *et al.*, 2021). However, the effectiveness of a blended mode of delivery for micro-credentials is subjective and requires more evidence to understand the impact on outcomes. Our research contributes to the ongoing debate and creates possibilities for extended research in the area.

2.3 Micro-credentials and employability

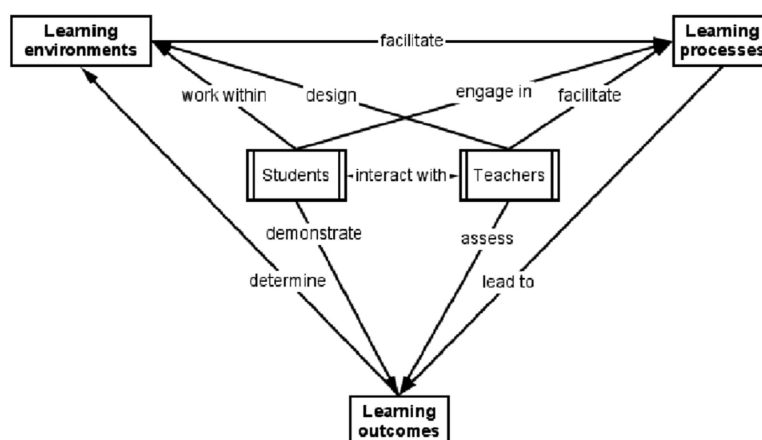
Robbins (1963) identified employability as one of the core objectives of higher education. Extending this further, Oria (2012) explained that employability includes transversal skills and competencies that contribute to leadership, communication, crisis management and creative thinking that are of national and international importance. Later, José Sá and Serpa (2018) explained that although some of the transversal skills have become a part of the curriculum and pedagogical practices, addressing them from a perspective that combines the national and international job market requirements is critical. Doing so can potentially support higher education institutions in fostering a productive learning environment and implementing new reward mechanisms.

Though the concept of employability is nothing new, scholars find it challenging to get a unified theory of employability (Hillage and Pollard, 1998). Therefore, multiple theoretical perspectives on employability exist in the literature. Theoretically, frameworks like the Understanding, Skills, Efficacy beliefs and Meta-cognition (USEM) framework (Knight and Yorke, 2002), CareerEDGE framework (Pool and Sewell, 2007) and LEPO framework (Phillips *et al.*, 2010) have been commonly used to define employability. While the studies using these frameworks or others have expansively explained graduate employability and transferability (Vasudeva and Mogaji, 2020; Kornelakis and Petrakaki, 2020) and employability and student representation at the workplace (Bonnard, 2020), recent studies have extended further to identify various in-demand employability skills and factors. However, the list of these in-demand skills is dynamic. According to Mohamad Zuber *et al.* (2020), employability skills are required in job readiness, where an individual is actively looking for a job. As understood, employability is a skill-driven phenomenon (Singh, 2020) and technological advancements, job market changes and education policies force stakeholders and policymakers to reimagine employability (Kohler *et al.*, 2021). Holidi and Seman (2023) found employability skills to combine fundamental skills and workplace competencies needed for a job.

Furthermore, studies have found that employability can be channeled through various factors, including micro-credentials (Tamoliune *et al.*, 2023). Research on micro-credentials has confirmed that micro-credentials can enhance employability (Orman *et al.*, 2023) and fill the skill gap in the job markets (Maina *et al.*, 2022). Therefore, higher education institutions must consider and integrate micro-credentials in their learning ecosystems. To study micro-credentials' role in enhancing employability, we postulate that micro-credentials and employability are related and use the LEPO framework to understand the interplay of the learning environment, learning processes and learning outcomes in higher education.

2.4 Theoretical framework - LEPO framework

The LEPO Framework (Figure 1), as proposed by Phillips *et al.* (2010), redefines learning using three components: (1) the learning environment (environment facilitating learning), (2) learning processes (activities) and (3) learning outcomes (knowledge, behavior, skills, understanding). This study conceptualizes the integration of micro-credentials in the higher education ecosystem within these three components, guided by different theoretical perspectives.



Source(s): Phillips *et al.* (2010)

Figure 1.
LEPO framework

Fraser (1998) defines learning environments as “social, psychological and pedagogical context that facilitates learning processes and affects the learning outcomes”. The most advocated theory for understanding learning environments is the theory of constructivism, which emphasizes facilitating the learning processes by constructing knowledge through previous experiences of learners (Kintsch, 1994). As a result, the learning environments must be designed, re-designed and updated to support the learning process, considering the learners’ social, cognitive and psychological context. This process leads to the construction of knowledge and the attainment of educational goals.

Another perspective emphasizes the sense of interdependency between the learning environment, learning process and learning outcomes. Garrison (2013) explains this interdependency regarding inquiry-based teaching and learning or experiential learning. An inquiry-based model supports that teaching and learning should encourage learners to self-reflect, learn and become critical thinkers. However, while most constructive approaches are limited to the learning environment and processes, Von Glasersfeld’s (1986) constructivism perspective also considers the learning outcomes. He states that learning is not limited to stimulus-response but requires self-regulation and development, leading to learning outcomes.

In the context of micro-credentials, the learning processes should absorb active learning principles and focus on intended learning goals (Biggs, 2011). Interactions, debates and discussions between the students, faculty and external experts are critical for higher engagement (Fryer *et al.*, 2021). According to the Humanist Theory of Learning, learning processes are subject to the learner’s needs and interests (Kurtz, 2000). Therefore, to build an ecosystem where micro-credentials can be embedded in the learning design of higher education, learning processes should be student-centric and the learning environment should be personalized.

Considering the different theoretical perspectives within constructivist ideology, we chose the LEPO framework to study how micro-credentials can be integrated into the higher education ecosystem. According to the LEPO framework, learning requires a systematic development of the learning environment, implementation of learning processes and diligent focus on the learning outcomes. It also emphasizes different interactions between the learners and faculty, which can be face-to-face and mediated through technology, postulating the blended approach. Even though LEPO has been extensively used to study student

preferences for traditional and electronic sources in medical education (Wong *et al.*, 2021), explore teaching and learning in multigrade classrooms (Msimanga, 2020) and understanding experiential education for achieving meaningful learning based on learners' self-experiences (Aswita, 2020), it can also be used to understand the transformations in learning environments and learning processes. It thus can also be used to evaluate learning outcomes by integrating micro-credentials to enhance employability. We explore and present this in the study through the following methodological approach.

3. Methodology

The study employs a qualitative methodology, recognizing its efficacy in delving into the nuanced experiences surrounding micro-credentials (Merriam and Tisdell, 2015). Following a similar approach by Sharma *et al.* (2022) and Jain *et al.* (2022) in understanding the multi-stakeholder perspective of relationship marketing in higher education institutions, our qualitative research methodology offers a rich exploration of the perspectives of learners, faculty, administrators and industry professionals, providing depth and context to the understanding of micro-credentials and their integration into the higher education ecosystem. Participants were purposively sampled to ensure a diverse representation of stakeholders. A total of 65 participants were engaged in the qualitative inquiry, comprising learners (44.6%, n = 29), faculty (26.1%, n = 17), administrators (15.3%, n = 10) and industry professionals (13.8%, n = 9), as presented in Table 1. The selection criteria included learners enrolled in higher education, faculty members, administrators with a basic understanding of micro-credentials and industry professionals involved in educational technology sectors. Participants were drawn from India (South Asia), Nigeria (sub-Saharan Africa), the United Arab Emirates (Asia/Middle East) and the United Kingdom (Western Europe) to capture a global perspective.

Online in-depth interviews, conducted over four months via Zoom, served as the primary data collection method. Facilitated by an interview guide (Appendixes A-1 and A-2), the interviews were video-recorded with participant consent. The choice of interviews allowed for a nuanced exploration of participants' perceptions and experiences regarding micro-credentials. Thematic analysis was employed for data analysis, drawing from Braun and Clarke's (2006) six phases. The process involved familiarizing myself with the data through repeated readings, immersing myself in understanding participants' perspectives and generating initial codes. Refinement of codes identified key themes, each carefully reviewed and categorized (Table 2). The credibility of the data was ensured through member checks (Merriam and Tisdell, 2015), affirming the authenticity and accuracy of the data. The research

Table 1.
Demographic profile of
participants in the four
categories

	Learners n = 29	Faculty n = 17	Administrators n = 10	Industry professionals n = 9
No. of Participants	India: 6 UK: 11 Dubai: 4 Nigeria: 8	India: 6 UK: 4 Dubai: 2 Nigeria: 5	India: 2 UK: 4 Dubai: 2 Nigeria: 2	India: 4 UK: 3 Nigeria: 2
Age Group	20–28	30–45	40–50	30–40
Gender	Female: 15 Male: 14	Female: 9 Male: 8	Female: 4 Male: 6	Female: 4 Male: 5
Work Experience	0–7 years	5–20 years	15–25 years	5–15 years
Highest Education	Secondary School: 15 Bachelor's Degree: 11 Master's Degree: 3	Master's: 2 Ph.D.: 15	Master's: 4 Ph.D.: 6	Master's: 7 Ph.D.: 2

Initial themes	Sub-themes	Main themes
1. Changes in higher education (n = 52) 2. Blended learning (n = 46) 3. Academic credit banks (n = 20) 4. Skill-based learning (n = 34) 5. Competency building (n = 22) 6. Digital learning (n = 58) 7. Online certifications (n = 32) 8. EdTech companies (n = 44) 9. Digital Infrastructure (n = 45) 10. Changing mindset (n = 23) 11. Flexibility of blended learning (n = 46) 12. Independent learning (n = 43) 13. Technology-mediated online and offline opportunity (n = 35) 14. New technological infrastructure (n = 40) 15. Technological cost (n = 32) 16. Long-term change (n = 57) 17. Preparedness (n = 55) 18. Open learning culture (n = 33) 19. Training for the new environment (n = 42)	Interventions and innovations Desirability for personal growth Advanced technical support Self-determination Technological acumen Risks involved Blended environment	<i>Theme 1: Micro-credentials in New Learning Ecosystem</i> “Learning is now redefined and digitized. In the context of the future, there will be a self-driven learning system. Freedom to make choices will govern our actions. My professors often inform us about this. We have discussions about how skills determine the job market. However, with an increase in the requirement for new skill sets, some skills might become obsolete but still will be an asset. I am currently pursuing two additional courses from an EdTech company. These will help me in presenting myself during the placements” (Student, Postgraduate) “Higher education is transforming. We are heading to a blended model of teaching and learning. With this the institutions are looking for new opportunities to improve their digital infrastructure and have more collaborations. This is a huge step taking us ahead and far from the traditional practices of learning. We in our institution are encouraging the same” (Faculty, PhD) <i>Theme 2: Micro-credentials and the Blended Learning Approach</i> “It is never easy to change the complete mindset overnight. Many countries across the world still do not have a robust education system. Forget that they are currently suffering from a crisis that has further exposed the vulnerabilities of the nations. There could be two possible outcomes; either this system will be completely neglected as people have to focus on their survival first, or the other way would be to take this as an opportunity to grow. The ones who may not be able to afford a full-time degree will possibly support it” (Industry professional, Postgraduate, UK) “Every change requires new skillsets, the pandemic has also obligated institutions to prepare for the worst. They need to prepare for a future which is uncertain. The only certainty is that technology is there to stay . . . right. Therefore, the institutions are now gearing up to invest in training programmes which can help them survive competition, adapt to new learning environment, and fulfill their goals” (Faculty, PhD, Nigeria)

(continued)

Table 2.
Coding table

Table 2.

Initial themes	Sub-themes	Main themes
20. Collaborations with industry (n = 61)	Academia-Industry partnership	<i>Theme 3: Micro-credentials and Learning Processes</i> “We have always talked about industry-academia collaboration. We invite our industry partners to take sessions for our students. This gives them exposure and creates possibility for employment. This is an important partnership between the two industries. Now, if I say that your EdTech companies are now a . . . an important part of this partnership, this implies that it is a multidimensional partnership” (Administrator, PhD, India)
21. Skill-based jobs (n = 41)		
22. Creativity (n = 21)		
23. New pedagogies (n = 26)		
24. Flipped classrooms (n = 17)	Advanced pedagogies	
25. Experiential learning (n = 15)	Student-centric learning	“Most of the learning happens outside the classroom. Students today learn from their peer or even prefer self-learning. This is empowering . . . I mean we now want our students to come and contribute during the sessions. I have a module where I ask students to take session. This is a game changer” (Faculty, PhD, Dubai)
26. Enjoyable learning experience (n = 24)		
27. Real-time learning (n = 22)	Student engagement and co-creation	
28. Global faculties (n = 43)		
29. Peer learning (n = 57)		
30. Involvement of students (n = 48)		
31. Self-paced learning (n = 38)	Increasing job opportunities	<i>Theme 4: Micro-credentials and Employability as a Learning Outcome</i> “The future is technology driven . . . I mean we have come to a point where being employed is no longer about physical settings. There are many opportunities online. This is what our system should prepare the students for. Taking new tasks and learning while working. This is holistic learning in true sense. All these will automatically lead to employability” (Administrator, PhD, India)
32. Increasing online job opportunities (n = 28)		
33. Entrepreneurship (n = 36)		
34. Content sharing (n = 27)		
35. Lifelong learning skills (n = 17)	Lifelong learning culture and practices	“Students, the ultimate aim is to earn. Online learning gives them the flexibility to learn while working. They can have their business and at the same time pursue a degree. Isn't that good. I think we should try to encourage this culture of credentialing and . . . it is flexible and easy. It creates more creative individuals” (Industry professional, Postgraduate, Nigeria)
36. Value to learning (n = 15)		
37. Stackable credentials (n = 18)		
38. Employability (n = 60)		
39. Independent growth (n = 33)		
40. Self-reinforcement (n = 49)		
Source(s): Author's own creation		

team maintained ethical considerations, emphasizing participants' voluntary participation and obtaining informed consent. The interaction between the research team and participants adhered to ethical standards, fostering a collaborative and transparent research environment. The diverse pool of participants and multi-stakeholder approach enhanced the study's robustness and contributed to a more comprehensive understanding of micro-credential experiences globally.

4. Findings

Upon analyzing the qualitative data from participant interviews, four distinct themes emerged, shedding light on the perspectives surrounding the micro-credentialing ecosystem in higher education. These thematic categories comprehensively understand the subject matter: 1. Micro-credentials in New Learning Environment, 2. Micro-credentials and the Blended Approach, 3. Micro-credentials and Learning Processes and 4. Micro-credentials and Employability as a Learning Outcome. To substantiate these findings, direct quotes from participants, each coded with a unique serial number, have been incorporated as evidence. This coding system ensures the traceability of participant perspectives and strengthens the reliability of the research findings. By employing this thematic approach supported by participant quotes, the analysis provides a nuanced and comprehensive exploration of the micro-credentialing ecosystem in higher education.

4.1 Theme 1: micro-credentials in new learning environment

The participants unanimously recognized the dynamic nature of the learning environment, acknowledging its variation across learners, who perceive it beyond traditional classrooms. In the realm of micro-credentials, the majority conveyed that the learning environment should extend beyond conventional settings, particularly when fostering lifelong learning. Creating a new learning environment was essential for integrating micro-credentials, breaking monotony and inspiring active engagement among learners and educators. This perspective aligns with [Valtonen et al.'s \(2020\)](#) notion of a learning environment focusing on space-technology pedagogy.

Further analysis of the data revealed that there are three enablers for integrating micro-credentials in the new learning environment:

4.1.1 External factors. 4.1.1.1 Rising demand for skill-driven education. Participants emphasized the increasing global demand for skill-driven education, driven by evolving job market needs. Micro-credentials were seen as a vital competency currency globally, catering to beginners and experienced professionals. A participant highlighted,

We are all witnessing transformations around us. Many factors contribute to these. The changes in the societal system, economic conditions, and, most importantly, the changes brought by the pandemic influence human choices. It is a realization that the future may demand survival of the fittest regarding skill set. We need to prepare for it -Administrator, Ph.D., Nigeria.

4.1.1.2 Partnerships and collaborations. The study identified partnerships between higher education, online learning platforms and government bodies as significant enablers. Collaborative programs with platforms like Futurelearn or Coursera were perceived to enhance the value of micro-credentials, offering flexibility, rich content and access to quality education. A University Manager from the UK stated that:

Partnering with online platforms like Futurelearn enhances the value of micro-credentials. This collaboration provides learners with flexibility, richer content, and access to quality education. Moreover, personalized courses align with industry needs, ensuring practical skills and a dynamic learning environment. - Administrator, Ph.D., UK

4.1.2 Personal factors. 4.1.2.1 Self-learning momentum. Participants stressed the importance of learners' self-learning abilities. 51 out of 65 participants affirmed that micro-credentials success relies on learners' profiles and capacity for independent learning. A student shared,

Learning is now redefined and digitized. In the context of the future, there will be a self-driven learning system. Freedom to make choices will govern our actions. My professors often inform us about this. We have discussions about how skills determine the job market. However, with an increase in the requirement for new skill sets, some skills might become obsolete but still be an asset. I am currently pursuing two additional courses from an EdTech company. These will help me present myself during the placements - Student, Postgraduate, U.K.

4.1.2.2 Motivation for continuous learning. A noteworthy personal factor contributing to micro-credentials success is the participants' motivation for continuous learning. The study revealed that learners who exhibited intrinsic motivation and a genuine interest in acquiring new skills were more likely to benefit from micro-credentials. Participants stressed that learners who view micro-credentials as opportunities for continuous skill enhancement rather than mere credentials are more likely to thrive in the evolving learning environment. For instance, a participant shared, "I believe in lifelong learning. Micro-credentials allow me to keep learning, updating my skills, and staying relevant in my field" - Student, Postgraduate, India.

4.1.3 Institutional factors. 4.1.3.1 Duration of programs. Longer-duration programs were valued more than short-duration ones. Participants indicated that longer durations contribute to better learning outcomes. Micro-credentials, often short-term, faced challenges as learners tend to value extended programs more, emphasizing the need for careful planning by institutions. One faculty member from UAE said:

As a faculty member, I've observed that students often perceive longer-duration programs as more valuable. The extended timeframe allows for a deeper exploration of topics and better knowledge retention. Micro-credentials, being short-term, encounter challenges as students prioritize the comprehensive learning experience offered by longer programs. Institutions must strategically plan the duration of micro-credentials to align with learners' expectations and enhance overall program efficacy - Faculty Member, Ph.D., UAE.

4.1.3.2 Standardization and nationwide acceptance. Standardizing micro-credentials and ensuring their nationwide acceptance emerged as critical enablers. Participants highlighted the barrier of lower acceptance compared to traditional degrees, emphasizing the importance of institutional collaboration with industry partners for standardization. An administrator from Nigeria shared her expectations:

As an administrator, I recognize the imperative of standardizing micro-credentials and securing nationwide acceptance. Participants see lower acceptance compared to traditional degrees as a significant hurdle. Institutional collaboration with industry partners becomes paramount for standardization, ensuring that micro-credentials align with industry expectations and seamlessly integrate into the broader credentialing system - Administrator, Masters, Nigeria.

In summary, all participant groups expressed a collective eagerness to enhance the learning environment with a focus on space-technology pedagogy. Learners, faculty, administrators and industry professionals emphasized the importance of a conducive learning environment for better engagement, content quality, mental and physical investments and improved outcomes. Additionally, identified personal, institutional and external enablers were deemed critical for developing a suitable learning environment, urging higher education institutions to consider these factors for the widespread integration of micro-credentials.

4.2 Theme 2: micro-credentials and the blended approach

Participants widely endorsed the blended approach as the optimal strategy for fostering the acceptance of micro-credentials, with 42 out of 65 expressing its high utility in higher

education integration. Despite their confidence, participants acknowledged the ongoing need for evidence, considering the developing nature of widespread micro-credential adoption. To emphasize the potential of the blended approach, a faculty member remarked:

Micro-credentials can be offered both online and offline. But a blend of both is very effective. We ask our students to complete a component of the subject online and the remaining during offline teaching. This increases their confidence and knowledge as they learn independently - Faculty, Ph.D., India.

Participants outlined two distinct possibilities in delving into the blended approach for integrating micro-credentials in higher education. These avenues offer nuanced perspectives on how formal and non-formal offerings can converge to enhance the acceptance and impact of micro-credentials. We explore each sub-theme and its potential implications in further detail.

4.2.1 A formal offering. 4.2.1.1 Hybrid credentialing. Integrating online short-term courses with offline classes for a section of a full-time course enhances engagement and value for micro-credentials. Collaboration with online learning platforms enables learners to take charge of their learning journey while staying connected with faculty for guidance.

4.2.1.2 Online credentialing. Designing and offering specific courses as advanced online programs, with industry partnerships, enhances micro-credentials credibility. Learners benefit from industrial training and exposure, reinforcing the practical applicability of acquired skills.

4.2.2 A non-formal offering. 4.2.2.1 Additional credentialing. Integrating micro-credentials into extra-curricular activities, such as hackathons, boot camp workshops and industry-partnered projects, enhances student morale and personality development.

4.2.2.2 Industry immersion programs. Embedding micro-credentials within immersive industry programs provides students with hands-on experience and exposure to real-world challenges. Students can engage in internships, project-based learning and workshops through collaborations with relevant industries. This approach enriches their skill set and fosters a deeper understanding of the practical applications of micro-credentials within specific professional contexts.

4.2.3 Benefits of the blended approach. Participants' recommendations align with the recognized benefits of a blended approach highlighted by scholars. [Pye et al. \(2015\)](#) discussed how it can increase student retention, while [Chen \(2022\)](#) emphasized its importance in enhancing the learning environment. Scholars like [Cortez \(2020\)](#) and [Hasanah and Malik \(2020\)](#) have acknowledged the blended approach as the new normal in higher education, further supporting the notion that it can serve as an enabler in increasing the importance of micro-credentials.

The blended approach saves time, encourages learners' self-discipline and opens avenues for working professionals to pursue courses for upskilling and re-skilling, ultimately advancing their careers. However, it is crucial to note that offering micro-credentials through a blended approach may be more feasible for postgraduate or executive programs, where a higher level of confidence, self-motivation and self-discipline tends to prevail.

4.3 Theme 3: micro-credentials and learning processes

The learning process, viewed as transformative and encompassing knowledge transfer between faculty and learners, emerged as a focal point in the participants' perspectives. In this study, participants underscored the pivotal role of various elements in driving the learning process, emphasizing its significance in competence building. In noting significant progress, administrators highlighted the collaborative efforts of faculty, staff and service providers who leverage advanced educational technology and active methodologies to enhance effectiveness.

4.3.1 Diverse perspectives on learning processes. 4.3.1.1 Faculty's holistic approach. Faculty members actively engage with industry experts and external guidance, contributing to a more holistic learning process. This collaborative approach ensures that the learning experience is not confined to academic realms but extends to real-world applications.

4.3.1.2 Learners' optimism. Students expressed optimism about the evolving learning process, particularly welcoming interdisciplinary approaches. They acknowledged the critical role of competence building, emphasizing the importance of cognition, intellect, skills and ability. The quality of the learning experience and the availability of alternative learning opportunities were deemed vital contributors to competence building.

4.3.1.3 Administrators' systemic view. Administrators provided a systemic view of higher education, highlighting the need for a comprehensive approach incorporating social, emotional and digital aspects. They emphasized the role of formal training, parallel industrial exposure and social awareness in creating a well-rounded learning environment.

4.3.2 Learning as a social practice. In the broader context of the discussion on learning processes within higher education, an illuminating perspective comes from an administrator based in Nigeria. This insightful quote underscores higher education's intricate and interconnected nature, portraying it not merely as an individual pursuit but as a comprehensive system that integrates various facets of competence. He said:

Higher education is not an individual entity. It is a system encapsulating different elements of competence. No institution can survive on only knowledge and skills. There must be a place for all social, emotional, or even digital levels. It is possible with formal training, parallel industrial exposure, and social awareness. It is a blend of all these; the micro-credentials offer these opportunities - Administrator, Postgraduate, Nigeria.

The administrator emphasizes that higher education extends beyond acquiring knowledge and skills, advocating for the inclusion of diverse dimensions—social, emotional and digital. The assertion that “Higher education is not an individual entity” reinforces that successful learning outcomes are contingent on a holistic approach. The administrator contends that for an educational institution to thrive, it must navigate beyond the traditional realms of academia, creating spaces for social interaction, emotional intelligence and digital proficiency. The quote posits that this multifaceted approach is achievable through formal training, parallel industrial exposure and social awareness. The administrator identifies micro-credentials as a catalyst for providing these diverse opportunities within the educational landscape and aligns with the evolving narrative that learning is a collaborative and social practice where the synergy between learners, faculty and technology becomes pivotal.

4.3.3 Technology's crucial role. Participants highlighted the fundamental role of technology and online platforms in the current learning process. This tripartite engagement—learner, faculty and technology—is essential for cultivating cognitive and social skills within the micro-credential context. Moreover, technology is seen as a tool to foster learner independence and promote lifelong learning, aligning with [Phillips et al. \(2010\)](#) explanation of learning processes driven by the learning environment and linked with meta-cognitive and self-regulatory skills. One student from UAE shared his perspective –

I believe technology is the backbone of our learning journey. In the UAE, the integration of online platforms has not only made learning more accessible but has also transformed the way we engage with educational content. The collaboration between learners, faculty, and technology has become the cornerstone of our micro-credential experiences, fostering academic growth and enhancing our cognitive and social skills. It's not just a tool; it's a gateway to independence in learning and a bridge to a lifelong educational journey. - Learner, UAE

4.3.4 Inference for effective learning processes. Inferences drawn from participants' insights emphasize that learning processes should be purposefully designed, driven by specific learning goals and tailored to integrate active learning. This approach aligns with the evolving landscape of micro-credentials, emphasizing the dynamic and collaborative nature of learning experiences.

4.4 Theme 4: micro-credentials and employability as a learning outcome

The participants, excluding learners, delved into the nuanced concept of employability, distinguishing it from mere job acquisition. They emphasized that employability is the capacity to perform effectively, while employment represents securing a job. In higher education, the role of institutions extends beyond facilitating employment to equipping individuals with the training and education necessary for career advancement.

Yorke and Knight (2006) underscored the significance of recognizing this distinction, advocating for the appreciation of employability as an achievement. The participants speculated on the positive relationship between micro-credentials and employability, viewing these as tangible indicators of an individual's readiness for a specific job.

Moreover, the participants underscored that employability is an essential outcome of the learning process. They acknowledged that both traditional and micro-credential credentials contribute to developing employability skills, ensuring individuals remain pertinent in the job market. However, they also cautioned that credentials alone, be they traditional degrees or micro-credentials, do not guarantee professional success. Continuous personal growth and performance are imperative for sustained professional development.

Two participant quotes further illuminated this perspective.

Degree reflects an individual's knowledge and attributes but does not guarantee expertise. As a result, they are inconsistent and may be discouraged by the employer. Micro-credentials can complement the degree and equip the individual with the project-based competency required for the job. However, completing the certification - Industry professional, Ph.D., India.

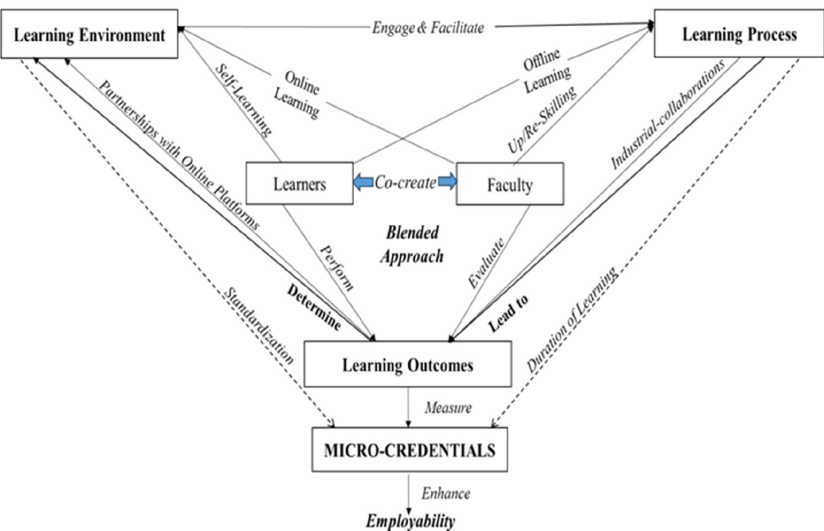
Higher education is not going to be the same anymore. It is an evolutionary cycle. We are heading to a future where employability will be defined by a student's skills and applying those skills to a given task. There is no escape. Upgrading and adapting to the change is the way. I will not say that degrees will lose their value, but even the universities need to change their model and realign with the new settings at work -Administrator, Ph.D., U.K.

Both quotes underscored the transformative potential of micro-credentials in bridging the gap between job market demands and the skill sets acquired through traditional education. These insights collectively convey the participants' belief that if widely standardized and accepted, micro-credentials can serve as formalized components within traditional degree systems. Participants envisioned a scenario where micro-credentials seamlessly integrate into higher education, allowing learners to accumulate credits and strategically leverage them in the job market—a concept they referred to as an academic bank of credit (A.B.C.). In their view, this holds the potential to empower learners in navigating the evolving landscape of employability and job acquisition. Based on these insights the following framework (Figure 2) was developed.

5. Discussion

The surge in educational technology and online learning platforms has propelled discussions on alternative teaching and learning approaches in higher education. Among these, micro-credentials have garnered attention, prompting an exploration of their integration into the higher education ecosystem. Leveraging Philip *et al.*'s LEPO framework, our study delves into the intricate relationship between micro-credentials, higher education and employability. While past studies using the LEPO framework primarily focused on student preferences,

Figure 2.
Framework for
learning ecosystem
driven by micro-
credentials for
enhancing
employability



Source(s): Authors' own creation

school environments and academic strategies, our adaptation extends the framework to contextualize micro-credentials within a technologically mediated learning environment—a blended approach.

Beyond expanding the LEPO framework, our study delves into the nuanced concept of employability, as distinguished by [Yorke and Knight \(2006\)](#). Recognizing employability as an achievement instrumental in job acquisition, we align our findings with the transformative efforts of higher education institutions. As articulated by [Erabaddage et al. \(2015\)](#) and [Singh and Tilak \(2020\)](#), institutions see micro-credentials as pivotal in augmenting employability by addressing skill gaps complementing traditional qualifications and underscoring micro-credentials' significance in shaping a student's professional profile, aligning with [Maina et al. \(2022\)](#) assertion that micro-credentialing positively reflects in a student's curriculum vitae.

Moreover, our study underscores the interdependence of the learning environment, learning process and learning outcomes in the context of micro-credentials. Parallel insights from [Xu and Zou \(2023\)](#) reinforce this relationship, particularly highlighting its significance during the pandemic. Emphasizing a blended approach as a suitable mode of delivery for micro-credentials, we draw on the flexibility, convenience and self-learning opportunities it offers ([Nikolopoulou and Zacharias, 2023](#)). However, the study invites future research to validate the correlation between the blended approach and micro-credential effectiveness through experimental or longitudinal studies.

Our study posits that micro-credentials wield the potential to shape learners' career trajectories when integrated into higher education institutions. Yet, this integration demands a visionary approach, technological parity, active learning models, national and industrial recognition and a transformative shift in mindset—a call for a comprehensive and collaborative effort to realize the full potential of micro-credentials in the contemporary learning landscape.

5.1 Theoretical contributions

5.1.1 Enriching the LEPO framework with micro-credentials integration. This study significantly enriches the LEPO Framework ([Phillips et al., 2010](#)) by applying it to integrating

micro-credentials in higher education. Building upon the theoretical perspectives within constructivist ideology and guided by [Fraser's \(1998\)](#) definition of learning environments, the findings demonstrate how micro-credentials reshape the learning environment by extending beyond traditional classrooms, aligning with the theory of constructivism. Additionally, the study identifies enablers for micro-credential integration within the LEPO Framework, incorporating external factors, partnerships, personal factors and institutional factors, which align with [Garrison's \(2013\)](#) idea of interdependency between the learning environment, learning process and learning outcomes.

5.1.2 Globalizing micro-credential scholarship. This research contributes to the globalization of micro-credential scholarship by addressing the prevailing focus on developed countries. By undertaking a multi-country analysis, the study extends the geographical scope of micro-credential research, aligning with the call for more diverse perspectives ([Ahsan et al., 2023](#)). The findings illuminate how micro-credentials are perceived in countries with distinct educational environments, challenges with technology and varied learner profiles, thereby broadening the understanding of the global dynamics of micro-credential adoption. Expanding the study's scope aligns with the LEPO Framework, emphasizing the need for a comprehensive approach incorporating social, emotional and digital aspects within different global contexts.

5.1.3 Blended learning as a catalyst for micro-credential integration. The study advances the theoretical understanding of the role of technology in micro-credential integration, contributing to the limited literature on the mode of delivery. Drawing on the LEPO Framework, the study highlights the blended approach as a strategic means of fostering the acceptance of micro-credentials in higher education and aligns with the Humanist Theory of Learning ([Kurtz, 2000](#)), emphasizing student-centric learning processes within personalized learning environments. By illustrating the benefits of a blended approach, the study addresses the gap in existing research on the delivery of micro-credentials ([Stefaniak and Carey, 2019](#)), showcasing how technology, faculty and learners interact to enhance cognitive and social skills within the micro-credential context.

5.2 Practical implications

The implications drawn from participant interviews offer actionable insights for key stakeholders in the micro-credentialing landscape.

5.2.1 University administrators. University administrators play a pivotal role in shaping the micro-credentialing landscape. To enhance the learning environment, administrators should implement a new space-technology-pedagogy-focused approach. Collaborating with online platforms and government bodies is crucial for personalized courses that align with industry needs. Supporting self-learning and intrinsic motivation requires strategic planning and innovative initiatives. Furthermore, administrators should partner with industry experts to standardize micro-credentials, ensuring nationwide acceptance. Embracing a blended approach in formal and non-formal program offerings can cater to diverse learning preferences and foster a dynamic learning ecosystem.

5.2.2 University lecturers and faculties. University lecturers are vital influencers in student learning experiences. They can contribute by adopting a blended approach, incorporating both online and offline components, encouraging independent learning and providing guidance during offline sessions. Advocating for the benefits of longer-duration programs and supporting blended credentialing can enhance micro-credentials' credibility. To adapt teaching methods to the evolving learning environment, continuous professional development is essential for lecturers. Collaboration with industry experts ensures course content's relevance and gives students real-world insights.

5.2.3 EdTech developers working on micro-credentials. EdTech developers hold a critical role in shaping the technology-driven future of micro-credentials. Developing user-friendly platforms that facilitate seamless learning experiences is essential. Offering diverse micro-credential programs with varying durations caters to the preferences of a diverse learner base. Platform collaboration features enable interactions between learners, faculty and industry experts. To enhance industry relevance, developers should ensure that micro-credentials address industry needs and provide practical skills. Exploring innovative technological integration, such as virtual or augmented reality, can enhance the learning experience.

5.2.4 Higher education policymakers. Higher education policymakers can influence the widespread acceptance of micro-credentials. Advocating for standardization ensures these credentials bridge the acceptance gap compared to traditional degrees. Policies supporting lifelong learning initiatives encourage continuous skill development. Facilitating collaborations between higher education institutions, online platforms and government bodies is crucial. Policymakers should develop frameworks promoting blended learning approaches, emphasizing the importance of online and offline components. Official recognition of micro-credentials within the broader credentialing system is crucial to their integration into higher education.

5.2.5 Students. Students are central to micro-credentials' success, actively engaging in the learning process. Embracing a mindset of lifelong learning is crucial, recognizing micro-credentials as opportunities for continuous skill enhancement. Exploring additional credentials integrated into extra-curricular activities enhances morale and personality development. Actively participating in blended learning programs making the most of online and offline components, contributes to a holistic learning experience. Developing a self-driven learning mindset, taking advantage of flexibility and accessibility, empowers students in navigating the evolving landscape of micro-credentials. Strategic planning of learning journeys, considering program duration and aligning with personal career goals ensure optimal student outcomes.

6. Conclusion

In conclusion, this study contributes to the LEPO framework and the theory of employability by showcasing the integration of micro-credentials in higher education through a blended delivery mode. However, certain limitations suggest avenues for future research. The ongoing debate on micro-credential standardization and the role of enablers in integration requires deeper exploration. While the study incorporates the European Commission's definition of micro-credentials into the LEPO framework, future researchers can leverage the unified theory of acceptance and use of technology (UTAUT) to delve into these enablers. Categorizing micro-credentials in future studies based on duration or type could provide more clarity and facilitate theory development. Expanding the research to other nations and investigating the impact of credential inflation would enrich the understanding of global micro-credentials. Further exploration of the advantages and disadvantages of the blended approach and a cross-country comparison between developed and developing nations could offer valuable insights into the nuances of micro-credential integration.

References

- Ahsan, K., Akbar, S., Kam, B. and Abdulrahman, M.D.A. (2023), "Implementation of micro-credentials in higher education: a systematic literature review", *Education and Information Technologies*, Vol. 28 No. 10, pp. 13505-13540, doi: [10.1007/s10639-023-11739-z](https://doi.org/10.1007/s10639-023-11739-z).
- Aswita, D. (2020), "Experiential education for meaningful learning: a literature study", *Proceedings of International Conference on Multidisciplinary Research*, Vol. 3 No. 2, pp. 83-92.

- Attri, R.K. (2019), *The Models of Skill Acquisition and Expertise Development: A Quick Reference of Summaries*, Speed To Proficiency Research: S2Pro©, Singapore.
- Biggs, J.B. (2011), *Teaching for Quality Learning at University: what the Student AISHE-J Volume 15, Number 1 (Spring 2023)*, 4th ed., Open University Press, Maidenhead, p. 16.
- Bonnard, C. (2020), "What is employability for higher education students?", *Journal of Education and Work*, Vol. 33 No. 5, pp. 425-445, doi: [10.1080/13639080.2020.1842866](https://doi.org/10.1080/13639080.2020.1842866).
- Bordoloi, R., Das, P. and Das, K. (2021), "Perception towards online/blended learning at the time of Covid-19 pandemic: academic analytics in the Indian context", *Asian Association of Open Universities Journal*, Vol. 16 No. 1, pp. 41-60, doi: [10.1108/aaouj-09-2020-0079](https://doi.org/10.1108/aaouj-09-2020-0079).
- Braun, V. and Clarke, V. (2006), "1,2. Using thematic analysis in psychology", *Qualitative Research in Psychology*, Vol. 3 No. 2, pp. 77-101, doi: [10.1191/1478088706qp063oa](https://doi.org/10.1191/1478088706qp063oa).
- Brown, M., Mhichil, M.N.G., Beirne, E. and Mac Lochlainn, C. (2021), "The global micro-credential landscape: charting a new credential ecology for lifelong learning", *Journal of Learning and Development*, Vol. 8 No. 2, pp. 228-254, doi: [10.56059/jl4d.v8i2.525](https://doi.org/10.56059/jl4d.v8i2.525).
- Chattopadhyay, A., Kupe, T. and Schatzer, N.F. (2022), "Fireside chat with three vice chancellors from three continents: Re-imagining higher education in emerging economies", in *Re-imagining Educational Futures in Developing Countries*, Palgrave Macmillan, Cham, doi: [10.1007/978-3-030-88234-1_5](https://doi.org/10.1007/978-3-030-88234-1_5).
- Chen, R.H. (2022), "Effects of deliberate practice on blended learning sustainability: a community of inquiry perspective", *Sustainability*, Vol. 14 No. 3, p. 1785, doi: [10.3390/su14031785](https://doi.org/10.3390/su14031785).
- Cirlan, E. and Loukkola, T. (2020), *Micro-credentials Linked to the Bologna Key Commitments*, 9-39, European University Association.
- Cortez, C.P. (2020), "Blended, distance, electronic and virtual-learning for the new normal of mathematics education: a senior high school student's perception", *European Journal of Interactive Multimedia and Education*, Vol. 1 No. 1, e02001, doi: [10.30935/ejimed/8276](https://doi.org/10.30935/ejimed/8276).
- Erabaddage, G., Shukri, M. and Khatibi, A. (2015), "Conceptualizing learning and employability learning and employability framework", *Journal of Education and Learning*, Vol. 4 No. 2, doi: [10.5539/jel.v4n2p53](https://doi.org/10.5539/jel.v4n2p53).
- Ericksen, K. (2019), "What are micro-credentials? How colleges are adapting to the modern workforce", available at: <https://collegiseducation.com/news/programs-and-course-content/what-are-micro-credentials/>
- Fraser, B.J. (1998), "Classroom environment instruments: development, validity and applications", *Journal of Learning Environment Research*, Vol. 1, pp. 7-34.
- Fryer, L.K., Shum, A., Lee, A. and Lau, P. (2021), "Mapping students' interest in a new domain: connecting prior knowledge, interest, and self-efficacy with interesting tasks and a lasting desire to reengage", *Learning and Instruction*, Vol. 75, 101493, doi: [10.1016/j.learninstruc.2021.101493](https://doi.org/10.1016/j.learninstruc.2021.101493).
- Gabriel, M. (2020), *A European Approach to Micro-credentials*, European Commission, European Union, p. 10.
- Garrison, D.R. (2013), "Theoretical foundations and epistemological insights", in Akyol, Z. and Garrison, D.R. (Eds), *Educational Communities of Inquiry: Theoretical Framework, Research, and Practice*, IGI Global, Hershey, PA.
- Gauthier, T. (2020), "The value of microcredentials: the employer's perspective", *Competency-based Education*, Vol. 5 No. 1209, pp. 1-6.
- Ghasia, M.A., Machumu, H.J. and DeSmet, E. (2019), "Micro-credentials in higher education institutions: an exploratory study of its place in Tanzania", *International Journal of Education and Development Using ICT*, Vol. 15 No. 1, pp. 1-12.
- Griffith, M. (2021), "How micro-credentials recognize 21st century skills and boost employability", in *Stakeholders of the Modern Credential Marketplace*, available at: <https://www.edalex.com/blog/how-educators-can-impact-employability-outcomes>

- Hasanah, H. and Malik, M.N. (2020), "Blended learning in improving students' critical thinking and communication skills at University", *Cypriot Journal of Educational Sciences*, Vol. 15 No. 5, pp. 1295-1306, doi: [10.18844/cjes.v15i5.5168](https://doi.org/10.18844/cjes.v15i5.5168).
- Healy, M. (2021), "Micro-credentials learners need quality careers and employability support", *The Journal of Teaching and Learning for Graduate Employability*, Vol. 12 No. 1, pp. 21-23, doi: [10.21153/jtlge2021vol12no1art1071](https://doi.org/10.21153/jtlge2021vol12no1art1071).
- Hillage, J. and Pollard, E. (1998), "Employability: developing a framework for policy analysis", *Labour Market Trends*, Vol. 4 No. 1, pp. 83-84.
- Holidi, N. and Seman, N. (2023), "A study of employability skills: from the employers and youth perspectives", *Journal of International Business Economics and Entrepreneurship*, Vol. 8 No. 1, pp. 64-76.
- Hunt, P.F. and Neves, A.M. (2020), *Practical Guide to Blended/Remote Learning and Children Disabilities*, UNICEF, New York, pp. 10-29.
- Ibrahim, M.M. and Nat, M. (2019), "Blended learning motivation model for instructors in higher education institutions", *International Journal Educational Technology in Higher Education*, Vol. 16 No. 1, p. 12, doi: [10.1186/s41239-019-0145-2](https://doi.org/10.1186/s41239-019-0145-2).
- Jain, V., Mogaji, E., Sharma, H. and Babbili, A.S. (2022), "A multi-stakeholder perspective of relationship marketing in higher education institutions", *Journal of Marketing for Higher Education*, pp. 1-19, doi: [10.1080/08841241.2022.2034201](https://doi.org/10.1080/08841241.2022.2034201).
- José Sá, M. and Serpa, S. (2018), "Transversal competences: their importance and learning processes by higher education students", *Education Sciences- MDPI*, Vol. 8 No. 126, pp. 1-12, doi: [10.3390/educsci8030126](https://doi.org/10.3390/educsci8030126).
- Kato, S. and Gyorfi, R. (2023), "Micro-credentials for lifelong learning and employability: uses and possibilities", OECD, Education Policies and Perspectives, No. 66, OECD Publishing, Paris.
- Kato, S., Galán-Muros, V. and Weko, T. (2020), *The Emergence of Alternative Credentials*, OECD Publishing, Paris.
- Kiiskilä, P., Kukkonen, A. and Pirkkalainen, H. (2023), "Are micro-credentials valuable for students? Perspective on verifiable digital credentials", *SN Computer Science*, Vol. 4 No. 4, pp. 1-16, doi: [10.1007/s42979-023-01797-y](https://doi.org/10.1007/s42979-023-01797-y).
- Kintsch, W. (1994), "Text comprehension, memory, and learning", *The American Psychologist*, Vol. 49 No. 4, pp. 294-303, doi: [10.1037/0003-066x.49.4.294](https://doi.org/10.1037/0003-066x.49.4.294).
- Knight, P.T. and Yorke, M. (2002), "Employability through the curriculum", *Tertiary Education and Management*, Vol. 8 No. 4, pp. 261-276, doi: [10.1080/13583883.2002.9967084](https://doi.org/10.1080/13583883.2002.9967084).
- Kohler, M., Gamrat, C., Raish, V. and Gross, E. (2021), *Microlearning and Micro-credentials in Higher Education*, Routledge, London.
- Kornelakis, A. and Petrakaki, D. (2020), "Embedding employability skills in U.K. higher education: between digitalization and marketization", *Industry and Higher Education*, Vol. 34 No. 5, pp. 290-297, doi: [10.1177/0950422220902978](https://doi.org/10.1177/0950422220902978).
- Kumar, J.A., Richard, R.J., Osman, S. and Lowrence, K. (2022), "Micro-credentials in leveraging emergency remote teaching: the relationship between novice users' insights and identity in Malaysia", *International Journal of Educational Technology in Higher Education*, Vol. 19 No. 1, p. 18, doi: [10.1186/s41239-022-00323-z](https://doi.org/10.1186/s41239-022-00323-z).
- Kurtz, P. (2000), *Humanist Manifesto 2000: A Call for a New Planetary Humanism*, Prometheus Books, Amherst, NY.
- Maina, M.F., Guàrdia Ortiz, L., Mancini, F. and Martinez Melo, M. (2022), "A micro-credentialing methodology for improved recognition of HE employability skills", *International Journal of Educational Technology in Higher Education*, Vol. 19 No. 1, pp. 1-22, doi: [10.1186/s41239-021-00315-5](https://doi.org/10.1186/s41239-021-00315-5).
- Matkin, G.W. (2020), "The challenge of digital credentials: how should universities respond?", in Burgos, D. (Ed.), *Radical Solutions and eLearning. Lecture Notes in Educational Technology*, Springer, Singapore.

- McGreal, R. and Olcott, D.A. (2022), "Strategic reset: micro-credentials for higher education leaders", *Smart Learning Environment*, Vol. 9 No. 9, pp. 1-23.
- McGreal, R., Mackintosh, W. and Cox, G.D.O.Jr (2022), "Bridging the gap: micro-credentials for development UNESCO chairs policy brief form. III world higher education conference (WHEC 2021) type: collective X", Vol. 23 No. 3, pp. 1-15.
- Merriam, S.B. and Tisdell, E.J. (2015), *Qualitative Research: A Guide to Design and Implementation*, the 4th ed., John Wiley & Sons, San Francisco, CA.
- Mischewski, B. (2017), "Micro-credentials: a model for engineering education", A Report Commissioned by the Tertiary Education Commission (T.E.C.), Engineering e2e.
- Mohamad Zuber, A.M., Hussin, M., Muhammad Helmi, N. and Saraswathy, K. (2020), "The employability skills among students of Public Higher Education Institutions in Malaysia".
- Msimanga, M. (2020), "Teaching and learning in multigrade classrooms: the LEPO framework", *Africa Education Review*, Vol. 17 No. 3, pp. 1-19, doi: [10.1080/18146627.2019.1671877](https://doi.org/10.1080/18146627.2019.1671877).
- Nikolopoulou, K. and Zacharis, G. (2023), "Blended learning in a higher education context: exploring university students' learning behavior", *Education Sciences*, Vol. 13 No. 5, p. 514, doi: [10.3390/educsci13050514](https://doi.org/10.3390/educsci13050514).
- Oliver, B. (2019), *Making Micro-credentials Work for Learners, Employers, and Providers*, Deakin University, Melbourne, available at: <http://dteach.deakin.edu.au/microcredentials>
- Oria, B. (2012), "Enhancing higher education students' employability: a Spanish case study", *International Journal of Technology Management and Sustainable Development*, Vol. 11 No. 3, pp. 217-230, doi: [10.1386/tmsd.11.3.217_1](https://doi.org/10.1386/tmsd.11.3.217_1).
- Orman, R., Şimşek, E. and Kozak Çakır, M.A. (2023), "Micro-credentials and reflections on higher education", *Higher Education Evaluation and Development*, Vol. 17 No. 2, pp. 96-112, doi: [10.1108/heed-08-2022-0028](https://doi.org/10.1108/heed-08-2022-0028).
- Phillips, R.A., McNaught, C. and Kennedy, G. (2010), "Towards a generalized conceptual framework for learning: the learning environment, learning processes and learning outcomes (LEPO) framework", In Herrington, J. and Hunter, W. (Eds), *ED-MEDIA*, 2495-2504.
- Pickard, L. (2018), "Analysis of 450 MOOC-based microcredentials reveals many options but little consistency", available at: <https://www.classcentral.com/report/moocs-microcredentials-analysis2018/>
- Pirkkalainen, H., Sood, I., Napoles, C.P., Kukkonen, A. and Camilleri, A. (2023), "How might micro-credentials influence institutions and empower learners in higher education?", *Educational Research*, Vol. 65 No. 1, pp. 40-63, doi: [10.1080/00131881.2022.2157302](https://doi.org/10.1080/00131881.2022.2157302).
- Pool, L.D. and Sewell, P. (2007), "The key to employability: developing a practical model of Education + Training", 49, 277-289.
- Pye, G., Holt, D., Salzman, S., Bellucci, E. and Lombardi, L. (2015), "Engaging diverse student audiences in contemporary blended learning environments in Australian higher business education: implications for Design and Practice", *Australasian Journal of Information Systems*, Vol. 19, pp. 1-20, doi: [10.3127/ajis.v19i0.1251](https://doi.org/10.3127/ajis.v19i0.1251).
- Robbins, L. (1963), *Higher Education Report of the Committee Appointed by the Prime Minister under the Chairmanship of Lord Robbins*, HMSO, London, pp. 1961-1963.
- Rossiter, D. and Tynan, B. (2019), "Designing and implementing micro-credentials: a guide for practitioners", *Commonwealth of Learning. The Knowledge Series*, available at: http://oasis.col.org/bitstream/handle/11599/3279/2019_KS_MicroCredentials.pdf?sequence=1&isAllowed=y
- Sankey, M.D. and Selvaratnam, R.M. (2021), "An integrative literature review of the implementation of micro-credentials in higher education: implications for practice in Australasia", *The Journal of Teaching and Learning for Graduate Employability*, Vol. 12, pp. 1-17, doi: [10.21153/jtlge2021vol12no1art942](https://doi.org/10.21153/jtlge2021vol12no1art942).
- Schick, S. (2021), "Everything you need to know about microcredentials", available at: <https://www.coursecompare.ca/what-are-microcredentials/>

- Sharma, H., Soetan, T., Farinloye, T., Mogaji, E. and Noite, M.D.F. (2022), "AI adoption in universities in emerging economies: prospects, challenges and recommendations", *Re-Imagining Educational Futures in Developing Countries: Lessons from Global Health Crises*, Springer International Publishing, Cham, pp. 159-174.
- Singh, S. (2020), "Employability challenges and issues in higher education", *Higher Education Review*, available at: <https://www.thehighereducationreview.com/opinion/in-my-view/employability-challenges-and-issues-in-higher-education-fid-99.html> (accessed 18 April 2021).
- Singh, D. and Tilak, G. (2020), "Implementation of education model 4.0: developing industry 4.0 skills in graduates engineers for improving employability skills", *Humanities and Social Sciences Reviews*, Vol. 8 No. 2, pp. 601-631, doi: [10.18510/hssr.2020.8268](https://doi.org/10.18510/hssr.2020.8268).
- Stefaniak, J. and Carey, K. (2019), "Instilling purpose and value in the implementation of digital badges in higher education", *International Journal of Educational Technology in Higher Education*, Vol. 16 No. 1, p. 44, doi: [10.1186/s41239-019-0175-9](https://doi.org/10.1186/s41239-019-0175-9).
- Tamolliune, G., Greenspon, R., Tereseviciene, M., Volungeviciene, A., Trepule, E. and Dauksiene, E. (2023), "Exploring the potential of micro-credentials: a systematic literature review", *Frontiers in Education*, Vol. 7, pp. 1-15, doi: [10.3389/feduc.2022.1006811](https://doi.org/10.3389/feduc.2022.1006811).
- Uggeri, M. and Barlassina, L. (2019), *Challenges and Opportunities of Micro-Credentials in Europe*, European Commission, Erasmus+, European Union.
- Valtonen, T., Leppänen, U., Hyypiä, M., Kokko, A., Manninen, J., Vartiainen, H., Sointu, E. and Hirsto, L. (2020), "Learning environments preferred by university students: a shift toward informal and flexible learning environments", *Learning Environments Research*, Vol. 24 No. 3, pp. 371-388, doi: [10.1007/s10984-020-09339-6](https://doi.org/10.1007/s10984-020-09339-6).
- Varadarajan, S., Koh, J.H.L. and Daniel, B.K. (2023), "A systematic review of the opportunities and challenges of micro-credentials for multiple stakeholders: learners, employers, higher education institutions, and government", *International Journal of Educational Technology in Higher Education*, Vol. 20 No. 13, pp. 1-24, doi: [10.1186/s41239-023-00381-x](https://doi.org/10.1186/s41239-023-00381-x).
- Vasudeva, S. and Mogaji, E. (2020), "Paving the way for world domination: analysis of African universities' mission statement", in Mogaji, E., Maringe, F. and Hinson, R.E. (Eds), *Understanding the Higher Education Market in Africa*, Routledge, Abingdon Oxfordshire.
- Von Glasersfeld, E. (1986), "Steps in the construction of 'others' and 'reality'", in Trapp, R. (Ed.), *Power, Autonomy, Utopia*, Plenum Press, London/New York, pp. 107-116.
- Wong, K., Kapoor, V., Tso, A., Oconnor, M., Convissar, D., Kothari, N. and Traba, C. (2021), "Environments, processes, and outcomes - using the LEPO framework to examine medical student learning preferences with traditional and electronic resources", *Medical Education Online*, Vol. 26 No. 1, pp. 8-23, doi: [10.1080/10872981.2021.1876316](https://doi.org/10.1080/10872981.2021.1876316).
- Xu, Y. and Zou, Y. (2023), "COVID-19 online teaching intervention and learning performance of college foreign language students", *Frontiers in Psychology*, Vol. 13, pp. 1-19, doi: [10.3389/fpsyg.2022.1109032](https://doi.org/10.3389/fpsyg.2022.1109032).
- Yorke, M. and Knight, P.T. (2006), *Embedding Employability into the Curriculum: Learning & Employability Series 1*, The Higher Education Academy, Innovation Way, York Science Park.
- Young, S., Deller, F. and McCallum, K.E. (2021), "Innovation in post-secondary education", in *Public Policy Forum*, Ottawa, available at: <https://fscf.ca/research/innovation-in-post-secondary-education/>

Further reading

- Adeyanju, S., Ajilore, O., Ogunlalu, O. and Onatunji, A. (2022), "Innovating in the face of the COVID-19 pandemic: case studies from Nigerian universities", in McKeown, J.S., Bista, K. and Chan, R.Y. (Eds), *Global Higher Education during COVID-19: Policy, Society, and Technology*, pp. 104-120.
- Braxton, S.N. (2023), "Competency frameworks, alternative credentials, and the evolving relationship of higher education and employers in recognizing skills and achievements", *International Journal of Information and Learning Technology*, Vol. 40 No. 5, pp. 373-387, doi: [10.1108/ijilt-10-2022-0206](https://doi.org/10.1108/ijilt-10-2022-0206).

-
- Chasi, S. (2022), "Re-Imagining international higher education partnerships in the aftermath of COVID-19", in *Re-imagining Educational Futures in Developing Countries*, Palgrave Macmillan, Cham, doi: [10.1007/978-3-030-88234-1_4](https://doi.org/10.1007/978-3-030-88234-1_4).
- Chemma, N., El Amine Abdelli, M. and Awasthi, A. (2022), "Management and information technology in the digital era: introduction to edited collection on challenges and perspectives", in Chemma, N., El Amine Abdelli, M. and Awasthi, A. (Eds), *Management and Information Technology in the Digital Era (Advanced Series in Management)*, Emerald Publishing, Bingley, Vol. 29, pp. 1-6.
- Clemons, S. (2019), *Assessing the Effectiveness of the Academic Assessing the Effectiveness of the Academic Strategies Course for Students on First Academic Strategies Course for Students on First Academic Probation Status Probation Status*, PhD Thesis, University Of Memphis Digital Commons, Tennessee.
- Daniela, D.P., Narce, G., Freddy, T.V.D.L., Nuria, L.R., Ariana, L., Antonio, P. and Sylvia, A. (2022), "Covid-19: study of online teaching, availability and use of technological resources", doi: [10.1007/978-3-030-88234-1_11](https://doi.org/10.1007/978-3-030-88234-1_11).
- Frank, A.B., Bartels, E.M. and Grissom, A.R. (2022), *Adaptive Engagement for Undergoverned Spaces (RR-A1275-1)*, Rand Corporation, California.
- Gupta, D.G. and Jain, V. (2022), "Influence of technology on student's integrated learning for effective well-being in developing countries", in Mogaji, E., Jain, V., Maringe, F. and Hinson, R.E. (Eds), *Re-imagining Educational Futures in Developing Countries*, Palgrave Macmillan, Cham.
- Kaur, J., Wadera, D. and Gupta, S. (2022), "Sustainable consumption practices in Indian households: a saga of environment management linked to Indian ethos and generational differences", *Society and Business Review*, Vol. 17 No. 3, pp. 441-468, doi: [10.1108/sbr-08-2021-0132](https://doi.org/10.1108/sbr-08-2021-0132).
- Ositelu, M.O., McCann, C. and Laitinen, A. (2021), "The short-term credentials landscape what we see and what remains unseen", available at: <https://www.newamerica.org/education-policy/reports/the-short-termcredentials-landscape/>

(The Appendix follows overleaf)

Table A1.
Focus areas and
discussion topics in the
interview protocol

Protocol segments	Respondent group	Retrieval (Challenges and disruptions)	Judgment (Collaboration and change)	Response (Competence)
Learners	Comprehension (communication)			
	• Understanding of the new normal in higher education	• How is the knowledge cycle evolving?	• What are the mechanisms for competency building?	• How will micro-credentials be embedded through blended learning to redefine higher education?
	• Micro-credentials	• What are the key enablers leading to the shift?	• How can micro-credentials be imbibed in the ecosystem?	
	• Blended Learning-	• Socio-cultural changes, technological advancements, global standards, change in job market	• Stakeholders' expectations	• What teaching and learning trends are there to stay post-COVID-19?
	• Pre- and post-COVID-19 learning		• Opportunities for collaboration and engagement	• Learning as an outcome of higher education, online programmes, peer connect, co-creation
Faculty	• Immersive learning environments			
	• Self-directed learning			
	• Upskilling			
	• How did micro-credentials evolve?	• What was the effect on students' understanding of competence and skill-building?	• What are the potential benefits of micro-credentials?	• How can institutions support and prepare for competency building?
	• What are some of the pros and cons of using a blended approach?	• Impact on classroom interaction, connectivity, student attention, and interest	• How will micro-credentials influence employability?	• How will lifelong learning fostered by micro-credentials redefine the scope of higher education?
Administrators	• Impact, issues and implications from pre-COVID, during COVID and post-COVID era			
	• How has micro-credential become integral to higher education?	• What administrative changes are expected to absorb the suggested changes?	• What could be some of the key challenges during these changes?	• How can institutions enhance scalability through the co-construction of credentials? What will lead to quality assurance in higher education?
		• How can educational credentials and micro-credentials support employability?		
Industry Professionals	• How important is skill-building for the future of work?	• What is the role of sectors in enhancing quality assurance and employability in higher education?	• How can constructions of micro-credentials infuse potential job opportunities?	• How will blended learning and micro-credentials redefine the landscape of higher education in the future?
	• How will micro-credentials supplement traditional qualifications to enhance employability?		• How do they think this change will be a personal and professional turning point in the given context of COVID-19?	• Future trends, collaborations, changing mindsets, and work culture
	• Preparing the students for industry, skill-based learning, self-driven	• Needs, opportunities, peer learning, technology access		

HIGHER EDUCATION 4.0:

1. How has higher education evolved as an ecosystem?
2. What have been the contributing factors to the emergence of education 4.0?

Probe: Technology, societal changes, economic changes.

3. How is education 4.0 contributing towards strengthening the goals of higher education?

Probe: Focus on employability, society, student experience, and research excellence.

MICRO-CREDENTIALS:

4. What is your understanding of micro-credentials?
5. What are some of the critical characteristics defining micro-credentials?

Probe: Short-term certifications, skill-building, micro-learning modules, alternative credentials.

6. How can micro-credentials be contextualized concerning professional development and skill-building?
7. How are they different from traditional degrees offered by the universities?
8. What are the enablers for the integration of micro-credentials in higher education?
9. What is the perceived value of micro-credentials in the job market?

Probe: How was the scenario before the advent of EdTech and micro-credentials?

NEP 2020: (Applicable only for participants from India)

10. What potential opportunities for higher education can be conceived from NEP 2020?

Probe: Discussion around NETF (National Educational Technology Forum).

11. What is the scope for micro-credentials in higher education?
12. How can the universities, government, and EdTech industry collaborate to support the new system of learning which integrates micro-credentials?

DRIVERS OF MICRO-CREDENTIALS:

13. What factors can ease the adoption of micro-credentials?

Probe: Technological, socio-economic, personal, professional, global, institutional, infrastructural, policy-related.

14. How do these apply in your institution/country?
15. What is the vision behind these enablers?
16. What are some of the critical challenges that persist in the adoption of micro-credentials?

EMPLOYABILITY:

17. How can employability be conceptualized in the 21st century?
18. How do micro-credentials act as enablers for enhancing employability?
19. How is the acceptance of static artifacts vs. live artifacts in the job market?

Probe: Static artifacts- College Degree and Live artifacts are- Certifications/ Credentials.

20. How will the new system help attain higher education quality, affordability, and equity?
21. What will be the role of universities, government, and the EdTech industry in promoting the culture of micro-credentials?

Probe: Independent roles and dependent roles.

GROWTH AND PORTABILITY (PRE-PANDEMIC, DURING, AND POST-PANDEMIC)

22. How has the partnership between universities, government, and the EdTech industry grown in the past decade? What have been the key contributing factors?
23. How has the pandemic influenced this tripartite relationship?
24. What have been some critical changes, and how will they affect the learning ecosystem in the future?
25. What kind of framework is needed to integrate micro-credentials as a part of higher education, facilitating portability for the learners and scalability for the universities?
26. How is your institution/country responding to this new change concerning micro-credentials?
27. What are your final thoughts or possible suggestions?

Corresponding author

Emmanuel Mogaji can be contacted at: e.mogaji@keele.ac.uk

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgroupublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com